

VibeGuard: Hardware Physical Measurement & Fit-Audit Guide

Project: VibeGuard — Edge AI Mechanical Vibration Diagnostic Testbed

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Target Hardware: N20 Micro Metal Gear Motor, Robocraze ADXL345 Breakout (#TJFKQXJUQ), M3 Fasteners, 12 mm Wooden Baseplate

Purpose: Precise physical dimensional audit mapping real hardware measurements to 3D-printed parts (FAB-01 Rotor Arm & FAB-02 Rigid Sensor Mount).

1. System Architecture: The Physical-to-CAD Connection

In precision additive manufacturing (FDM 3D printing in PLA), nominal CAD models must account for real hardware manufacturing tolerances and material thermal shrinkage. The following table establishes the direct causal connection between physical dimensions and CAD features:

Physical Measurement	Physical Component Feature	Affected 3D Print Part	Target CAD Feature in Model	Functional Impact
D_{shaft}	N20 D-Shaft Outer Diameter	FAB-01 Rotor Arm	Hub D-Bore Diameter	Snug press-fit on shaft without radial play or splitting
T_{flat}	N20 D-Shaft Flat Thickness	FAB-01 Rotor Arm	Hub Internal Flat Chord	Anti-slip torque lock preventing rotor arm free-spinning
L_{shaft}	N20 Usable Shaft Length	FAB-01 Rotor Arm	Hub Internal Bore Depth	Ensures 1 mm air gap from brass gearbox face

Physical Measurement	Physical Component Feature	Affected 3D Print Part	Target CAD Feature in Model	Functional Impact
				(prevents friction drag)
D_{washer}	M3 Steel Washer Outer Diameter	FAB-01 Mass Boss & Baseplate	Riser Pedestal Height (15 mm)	Rotational sweep radius clearance; guarantees +6.5 mm daylight gap
P_{hole}	ADXL345 PCB Mounting Hole Pitch	FAB-02 Sensor Mount	Upright Wall Dual M3 Hole Pitch	Dual bolt alignment; prevents tilted PCB mounting and axis cross-talk
D_{pcb\hole}	ADXL345 PCB Hole Diameter	FAB-02 Sensor Mount	Upright Hole Clearance (3.2 mm)	M3 screw pass-through without binding
T_{solder}	ADXL345 Backside Solder Blobs	FAB-02 Sensor Mount	Upright Face Relief Cavity	Flush planar seating against bracket wall (prevents 5°–10° tilt)
P_{bracket}	N20 Aluminum U-Bracket Pitch	Baseplate & Riser	Riser Through-Hole Spacing (17 mm)	Direct through-bolting through U-bracket and riser into wood
T_{plank}	Baseplate	Baseplate &	M3 Mounting	Ensures screw

Physical Measurement	Physical Component Feature	Affected 3D Print Part	Target CAD Feature in Model	Functional Impact
	Actual Thickness	Feet	Screw Length Selection	threads grip wood without poking out the bottom

2. Exhaustive Caliper Measurement Protocol

Use a digital or vernier caliper set to metric (0.01 mm or 0.05 mm resolution).

[CALIPER OUTSIDE JAWS]	[CALIPER DEPTH ROD]
[==== COMPONENT ====]	[=====] === (Depth)

2.1 N20 Micro Metal Gear Motor

Measurement 1: Shaft Round Outer Diameter (D_{shaft})

- **Where to place jaws:** Across the cylindrical, curved section of the steel output shaft.
- **How to measure:** Clamp the wide outside jaws perpendicularly across the shaft. Take 2 readings at 90° to check circularity.
- **Nominal:** 3.00 mm (typical range: 2.95–3.02 mm).
- **CAD Connection:** FDM PLA shrinks inward by ~ 0.15 mm inside small cylindrical bores. If your measured shaft is 2.98 mm, the CAD bore is modeled at 3.15 mm to achieve a perfect push-fit.

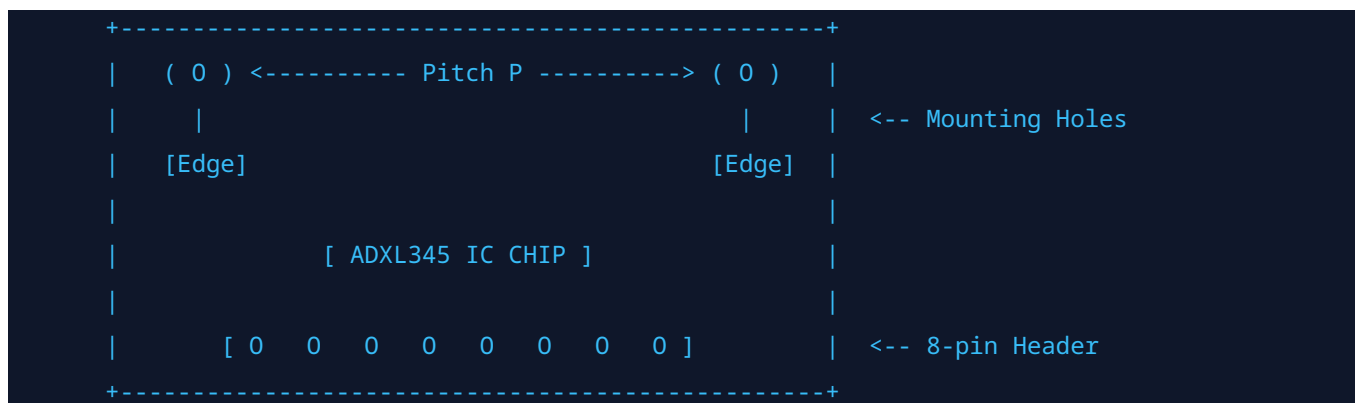
Measurement 2: Shaft Flat-to-Back Thickness (T_{flat})

- **Where to place jaws:** Across the D-cut profile.
- **How to measure:** Place one caliper flat jaw directly on the flat milled face of the D-shaft and the opposing jaw against the curved back.
- **Nominal:** 2.50 mm (cut depth is 0.50 mm; typical range: 2.42–2.55 mm).
- **CAD Connection:** The chord height inside [FAB-01](#) must be modeled as $T_{\text{flat}} + 0.125$ mm (2.625 mm). If too thick, the arm wobbles; if too thin, it cannot be pressed on.

Measurement 3: Shaft Usable Length (L_{shaft})

- **Where to place probe:** From the tip of the D-shaft back to the brass bearing collar on the gearbox front plate.
- **How to measure:** Extend the caliper depth rod out the bottom. Rest the caliper base against the shaft tip and slide the depth rod until it touches the front brass face.
- **Nominal:** 9.5–10.0 mm.
- **CAD Connection:** Hub depth is modeled at $L_{\text{shaft}} - 1.0 \text{ mm}$ (8.5 mm). This ensures the plastic never rubs against the rotating brass gearbox casing.

2.2 Robocraze ADXL345 Accelerometer Breakout Board



Measurement 4: PCB Mounting Hole Diameter ($D_{\text{pcb_hole}}$)

- **Where to place jaws:** Inside either of the two copper-ringed mounting holes.
- **How to measure:** Use the upper, pointed **inside jaws**. Insert them into the hole and expand gently until both knife edges touch the copper rim.
- **Nominal:** 3.1–3.2 mm (clearance for M3).
- **CAD Connection:** Verifies that M3 machine screws pass through without binding. If a board has 2.6 mm holes, it requires M2.5 screws.

Measurement 5: Center-to-Center Hole Pitch (P_{hole})

- **Where to place jaws:** Between the two mounting holes.
- **How to measure (The Pro Caliper Trick):**

1. Measure the diameter of Hole 1: D.
2. Measure from the **outer left rim** of Hole 1 to the **outer right rim** of Hole 2: W_{outer}.

3. Calculate pitch: $P = W_{\text{outer}} - D$.

(Alternatively: Measure from the left inner rim of Hole 1 to the left inner rim of Hole 2 directly).

- **Nominal:** 15.0 mm (or 15.24 mm / 0.6 inch grid).
- **CAD Connection:** Dictates the exact center-to-center distance between the two horizontal M3 through-holes on the vertical wall of [FAB-02](#). A 0.5 mm error will wedge the screws and stress the PCB.

Measurement 6: Backside Pin Solder Protrusion (T_{solder})

- **Where to place probe:** On the rear surface of the PCB behind the soldered 8-pin header.
 - **How to measure:** Measure how far the clipped pin ends or solder joints protrude beyond the flat fiberglass surface.
 - **Nominal:** 1.0–1.8 mm.
 - **CAD Connection:** If pins protrude, the vertical mounting wall of [FAB-02](#) incorporates a clearance pocket so the PCB sits 100% flush and orthogonal without tilting.
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2.3 Fasteners & Unbalance Counterweight

Measurement 7: Washer Outer Diameter (D_{washer})

- **Where to place jaws:** Across the diameter of the steel washers used on the M3 unbalance bolt.
- **How to measure:** Clamp caliper outside jaws across the widest diameter of the washer.
- **Nominal:** Standard DIN 125 is 7.0 mm; Wide DIN 9021 is 9.0 mm.
- **CAD Connection:** Directly determines the **Rotational Sweep Radius:**

$$R_{\text{sweep}} = 12.0 \text{ mm} + \frac{D_{\text{washer}}}{2}$$

At $D = 9.0 \text{ mm}$, $R_{\text{sweep}} = 16.5 \text{ mm}$. With our 15 mm riser, the shaft sits at $Y = 35.0 \text{ mm}$, yielding +6.5 mm daylight clearance above the 12 mm baseplate.

3. Caliper Audit Checklist

Copy and record your exact readings below:

Component	Target Feature	Nominal (mm)	Your Caliper Reading (mm)	Difference (\Delta)
N20 Motor	Shaft Round OD (D_{shaft})	3.00	_____	_____
N20 Motor	Shaft Flat Thickness (T_{flat})	2.50	_____	_____
N20 Motor	Usable Shaft Length (L_{shaft})	9.50	_____	_____
N20 Motor	Brass Collar Diameter	4.20	_____	_____
N20 U-Bracket	Base Mounting Hole Pitch	17.00	_____	_____
N20 U-Bracket	Base Plate Thickness	2.00	_____	_____
ADXL345 Board	PCB Hole Diameter	3.20	_____	_____
ADXL345 Board	Hole Center-to-Center Pitch	15.00	_____	_____
ADXL345 Board	Backside Solder Protrusion	1.20	_____	_____

Component	Target Feature	Nominal (mm)	Your Caliper Reading (mm)	Difference (\Delta)
M3 Fasteners	Washer Outer Diameter (D_{washer})	9.00	_____	_____
M3 Fasteners	M3 Bolt Under-Head Length	15.00	_____	_____
Wood Plank	Actual Baseplate Thickness	12.00	_____	_____

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